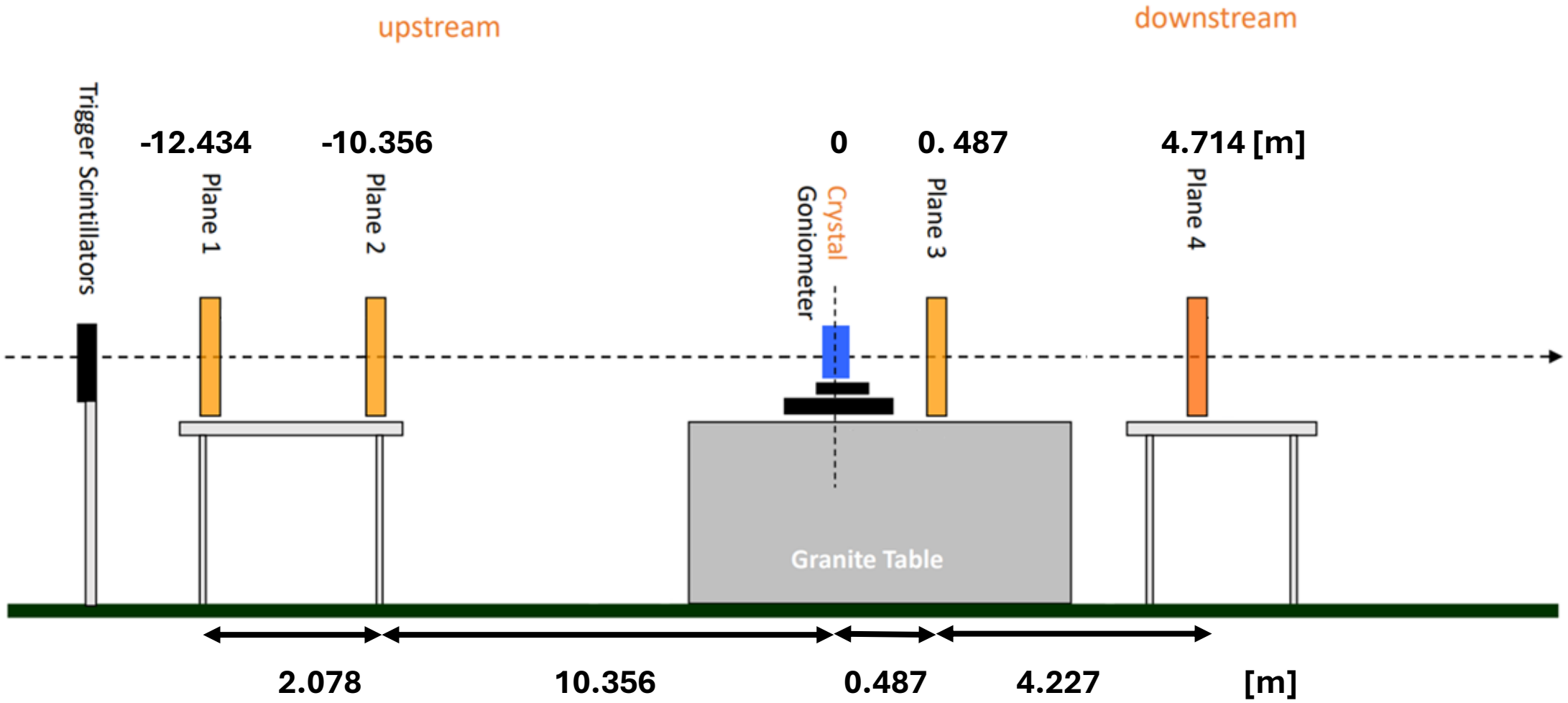
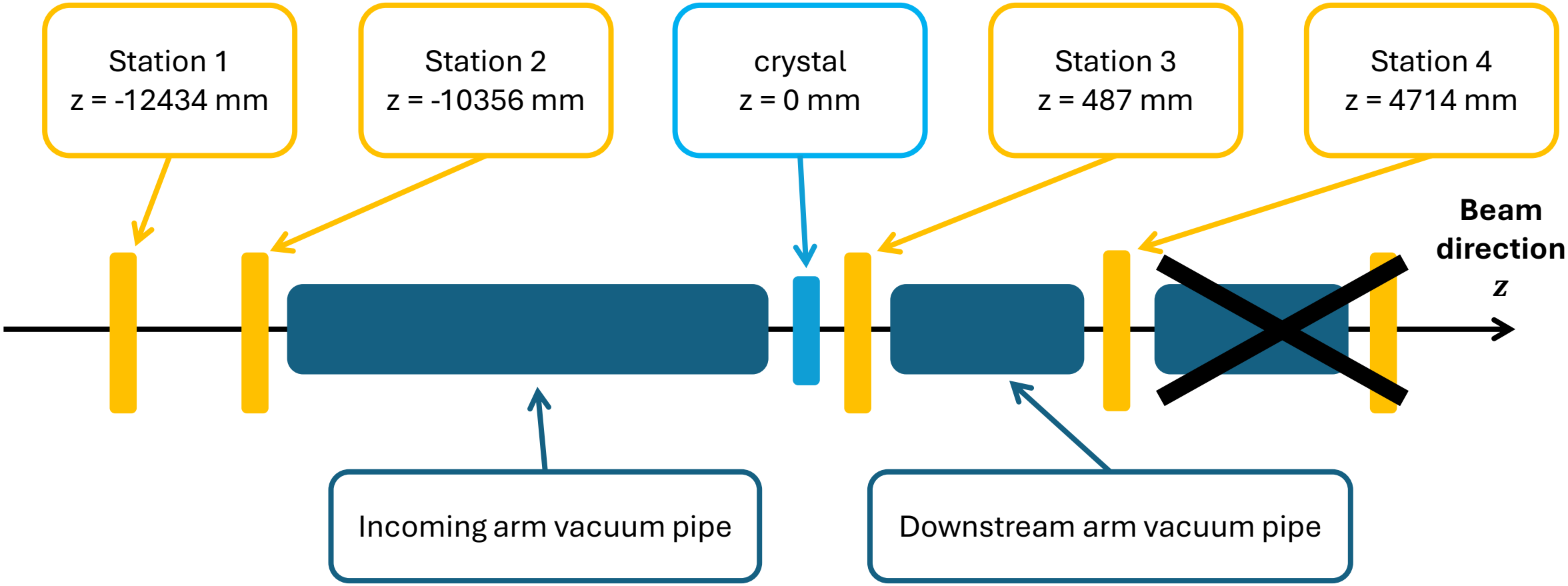
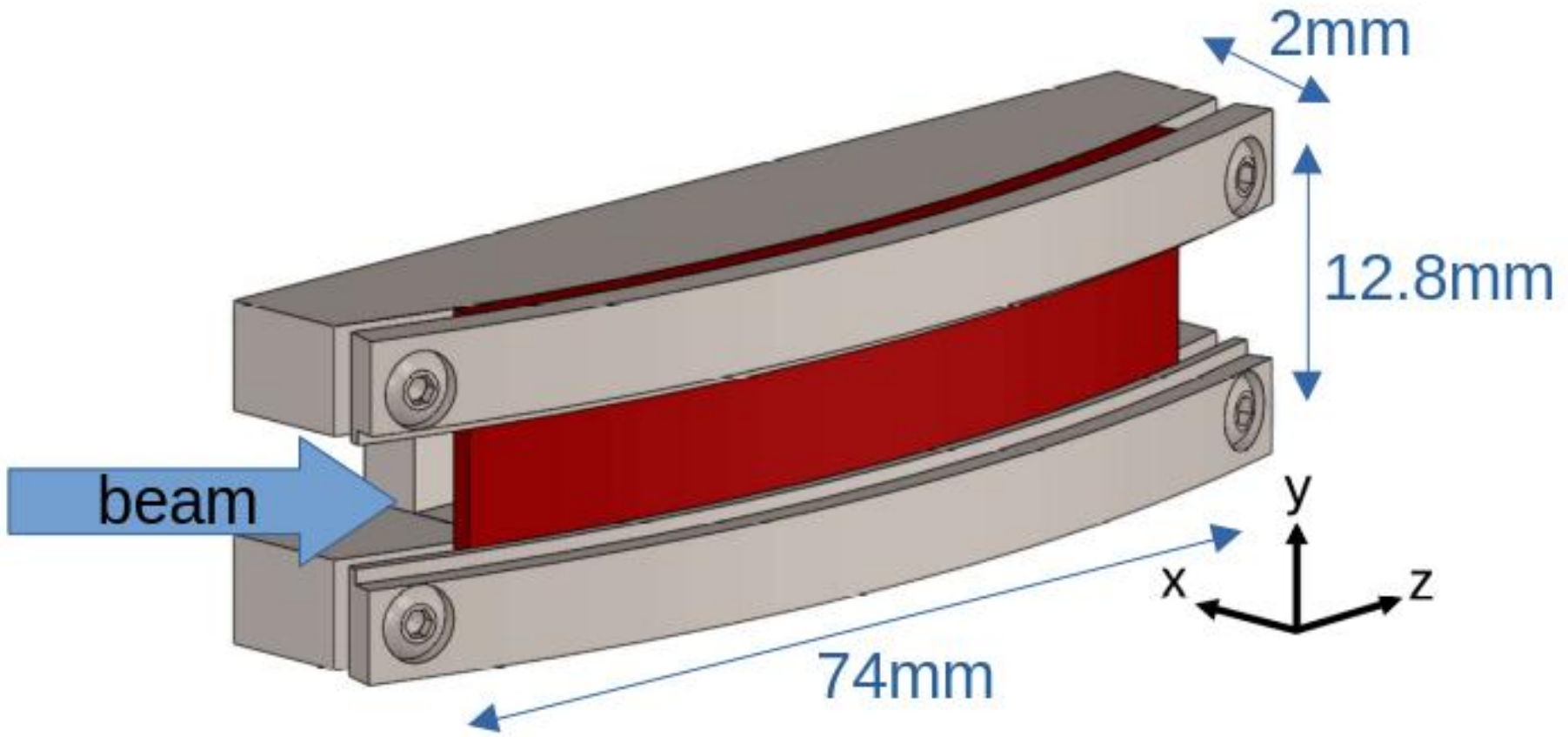
SCHEMATIC LAYOUT



SCHEMATIC LAYOUT



SCHEMATIC LAYOUT



VACUUM PIPES

- Vacuum pipes segments fill the arm specifically to suppress MCS in air between stations, the tracker stations themselves sit in the short breaks where a pipe segment has been removed to house the plane
- In a standard UA9 5-plane layout, the longest pipe sections are between plane 1-2 and 4-5
- For large bending long crystals, only 4 planes are used, and the downstream is deliberately short

BEAM FEATURES

- H8 secondary beamline off the SPS, driven by 400 GeV protons.
- Beamline operates with a target, and a spectrometer selection 180 GeV hadrons (mixed 30% π^+ and 70% p)
  - Target material: beryllium
  - Target thickness
- Beam divergence: order tens of urad

TRACKERS FEATURES

- 4 silicon detectors, 2 upstream and 2 downstream used to track
- Each station: two silicon microstrip sensors (each measures position in one dimension)
- Each sensor:
  - Active length 98 mm
  - 639 strips on a 60 um pitch and … um width
  - 38 x 38 mm^2 aperture
  - ~ 7 um resolution
  - Nominal thickness 320 $\pm$ 20 um
  - Measured thickness 328 $\pm$ 2 um